

Change the title of Clause 170 (added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024) as follows:

170. Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s (800GMII) and 1.6 Tb/s (1.6TMII)

170.1 Overview

Change the text in 170.1 as follows:

This clause defines the characteristics of the Reconciliation Sublayer (RS) and ~~the 800 Gb/s Media Independent Interface (800GMII) between Ethernet media access controllers and various 800 Gb/s PHYs for 800 Gb/s and 1.6 Tb/s PHYs.~~ Figure 170–1 shows the relationship of the RS and ~~800GMII to the ISO/IEC OSI reference model.~~ Media Independent Interface to the ISO/IEC OSI reference model. Note that there are two variants of the Media Independent Interface defined in this clause, the 800 Gb/s Media Independent Interface (800GMII) and the 1.6 Tb/s Media Independent Interface (1.6TMII).

The 800GMII and 1.6TMII ~~are~~ is an optional logical interfaces between the MAC sublayer and the Physical Layer device (PHY). The 800GMII/1.6TMII Extender may optionally be used to extend the 800GMII/1.6TMII (see Clause 171).

The RS adapts the bit serial protocols of the MAC to the parallel format of the PCS service interface. Though the 800GMII and 1.6TMII ~~is an~~ are optional interfaces, ~~it is~~ they are used in this standard as a basis for specification. The Physical Coding Sublayer (PCS) is specified to the 800GMII/1.6TMII, so if not implemented, a conforming implementation shall behave functionally as if the RS and 800GMII/1.6TMII were implemented.

~~The 800GMII has the following characteristics:~~

- ~~a) It supports a speed of 800 Gb/s.~~
- ~~b) Data and delimiters are synchronous to a clock reference.~~
- ~~c) It provides independent 64-bit wide transmit and receive data paths.~~
- ~~d) It supports full duplex operation only.~~

The 800GMII/1.6TMII have the following characteristics:

- The 800GMII supports a speed of 800 Gb/s.
- The 1.6TMII supports a speed of 1.6 Tb/s.
- Data and delimiters are synchronous to a clock reference.
- They provide independent 64-bit wide transmit and receive data paths.
- They support full duplex operation only.

Replace Figure 170–1 with the following figure:

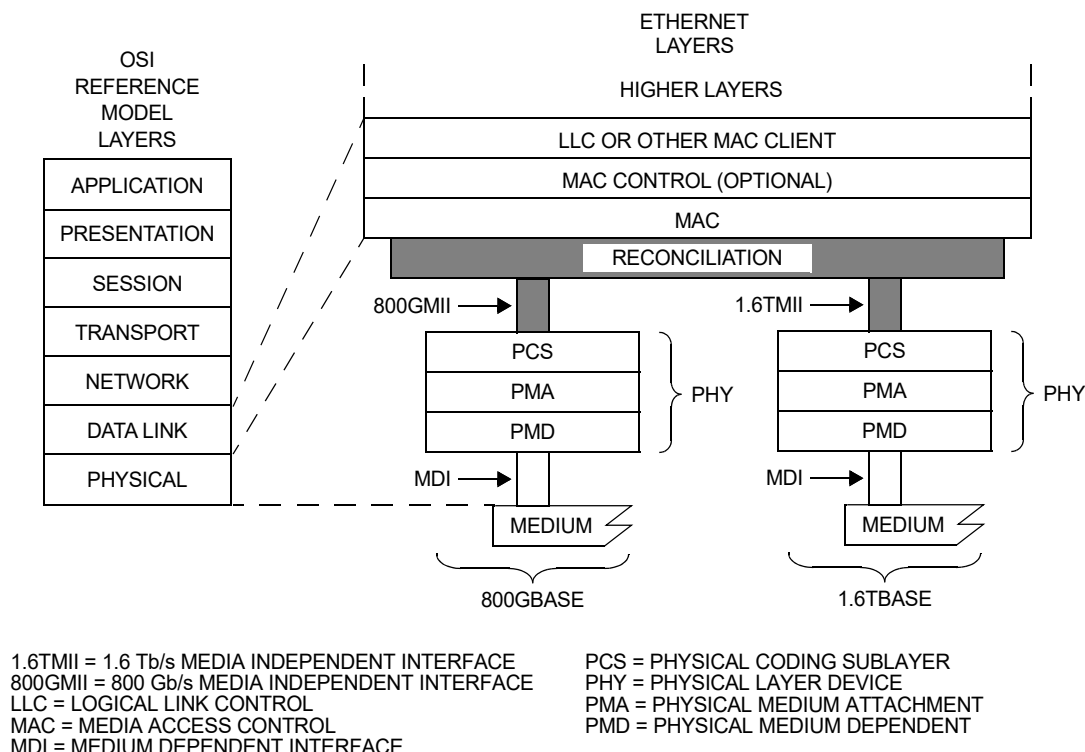


Figure 170–1—RS, 800GMII, and 1.6TMII relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

170.1.1 Summary of major concepts

Change the text in 170.1.1 as follows:

The following are the major concepts of the 800GMII/1.6TMII and RS:

- The 800GMII/1.6TMII are is-functionally similar to other media independent interfaces that have been defined for lower speeds, as they all define an interface allowing independent development of MAC and PHY logic.
- The RS converts between the MAC serial data stream and the parallel data paths of the 800GMII/1.6TMII.
- The RS maps the signal set provided at the 800GMII/1.6TMII to the PLS service primitives provided at the MAC.
- Each direction of data transfer is independent and serviced by data, control, and clock signals.
- The RS generates continuous data or control characters on the transmit path and expects continuous data or control characters on the receive path.

- The RS participates in link fault detection and reporting by monitoring the receive path for status reports that indicate an unreliable link, and generating status reports on the transmit path to report detected link faults to the DTE on the remote end of the connecting link.

170.1.2 Application

Change the text in 170.1.2 as follows:

~~This clause applies to the interface between the MAC and PHY, or between the PHY 800GXS and the PHY when the optional 800GMII Extender is used. This clause applies to the interface between the MAC and PHY. An 800GMII Extender or 1.6TMII Extender (see Clause 171) may be used to extend the reach of the 800GMII or 1.6TMII, respectively.~~ This logical interface is used to provide media independence so that an identical media access controller may be used with supported PHY types.

170.1.3 Rate of operation

Change the text in 170.1.3 as follows:

The 800GMII is specified to support 800 Gb/s operation. The 1.6TMII is specified to support 1.6 Tb/s operation.

170.1.4 Delay constraints

Change the text and table in 170.1.4 as follows:

The maximum cumulative MAC Control, MAC, and RS delay (sum of transmit and receive delays at one end of the link) shall meet the values specified in Table 170–1. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references for 800 Gb/s, and in 174.4 and its references for 1.6 Tb/s.

Table 170–1—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
800 Gb/s MAC, RS, and MAC Control	196 608	384	245.76
<u>1.6 Tb/s MAC, RS, and MAC Control</u>	<u>393 216</u>	<u>768</u>	<u>245.76</u>

170.1.5 Allocation of functions

Change the text in 170.1.5 as follows:

The allocation of functions at the 800GMII/1.6TMII balances the need for media independence with interface simplicity. The 800GMII/1.6TMII provides media independence by separating the Data Link and Physical Layers of the OSI seven-layer reference model.

Change the title and text of 170.1.6 as follows:

170.1.6 800GMII/1.6TMII structure

The 800GMII/1.6TMII structure is identical to the CGMII structure specified in 81.1.6.

Change the title and text of 170.1.7 as follows:

170.1.7 Mapping of 800GMII/1.6TMII signals to PLS service primitives

The Reconciliation Sublayer (RS) shall map the signals provided at the 800GMII/1.6TMII to the PLS service primitives defined in Clause 6. The PLS service primitives provided by the RS and described here behave in exactly the same manner as defined in Clause 6. Full duplex operation only is implemented at 800 Gb/s, therefore, PLS service primitives supporting CSMA/CD operation are not mapped through the RS to the 800GMII/1.6TMII. This behavior and restrictions are the same as described in 22.2.1, with the details of the signaling described in 170.3.

The RS maps all primitives in an identical manner as the CGMII does and as specified in 81.1.7 except that EEE and LPI signaling are not supported.

Change the title and text of 170.2 as follows:

170.2 800GMII/1.6TMII data stream

The 800GMII/1.6TMII data stream has the same characteristics as the CGMII data stream described in 81.2.

Change the title and text of 170.3 as follows:

170.3 800GMII/1.6TMII functional specifications

The 800GMII/1.6TMII functions identically to the CGMII specified in 81.3 except that EEE and LPI signaling are not supported.

Change the title of 170.4 as follows:

**170.4 Protocol implementation conformance statement (PICS) proforma for
Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for
800 Gb/s and 1.6 Tb/s operation (800GMII and 1.6TMII)²**

170.4.1 Introduction

Change the text in 170.4.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s and 1.6 Tb/s operation (800GMII and 1.6TMII), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

170.4.2 Identification

170.4.2.2 Protocol summary

Change the table in 170.4.2.2 as follows:

Identification of protocol standard	IEEE Std 802.3df-202x IEEE Std 802.3dj-202x, Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s <u>and 1.6 Tb/s</u> operation (800GMII <u>and 1.6TMII</u>)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-202x IEEE Std 802.3dj-202x.)	

Date of Statement	
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²Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

170.4.3 Major capabilities/options

Change the table in 170.4.3 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*PHY800	PHY support of 800GMII	170.2, 170.3		<u>O</u>	<u>Yes</u> [] <u>No</u> []
*PHY1600	PHY support of 1.6TMII	170.2, 170.3		<u>O</u>	<u>Yes</u> [] <u>No</u> []
*MII	PHY-Reconciliation Sublayer support of <u>either 800GMII or 1.6TMII</u>	170.2, 170.3		O	Yes [] No []

Change the title of 170.4.4 as follows:

170.4.4 PICS proforma tables for Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s and 1.6 Tb/s operation (800GMII and 1.6TMII)

170.4.4.3 Data stream structure

Change the table in 170.4.4.3 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
DS1	Frame transfer	170.2	Within 800GMII/ <u>1.6TMII</u> data stream	MII:M	Yes [] N/A []
...					

Change the title of 170.4.4.4 as follows:

170.4.4.4 800GMII/1.6TMII signal functional specifications